

Bound clock: evidence and limits

Signal Space / GROSS Test 6

Technical completion: completed; bounded scientific classification: pass.

Run run-080a63d117abd84d; analysis analysis-0001-2639af35.

Solver revision 8c42249f03b447e814c9d8017ce2cc02c0484eed; renderer revision 8c42249f03b447e814c9d8017ce2cc02c0484eed.

Flat decoupling limit of SS OCF 1, $c=\hbar=m=1$; constant internal doublet direction, $a=0$, gravity decoupled.

Four trial core frequencies 0.868, 0.875, 0.90, 0.94. Independently solved bound spectrum and trapping-off control.

Nominal peak $\chi=0.001$ for 100 eigenperiods; zero, half and double controls for 20 periods; independent $h/2$ and $2R$ controls for 20 periods.

Selected branch: 0.900. No emergent geometry, Test 7 reception or nonspherical stability established.

core-and-well

Question: Which trial core supplies a spatial mass well and localized clock eigenfunction?

Reading: Selected $\omega_Q=0.900$; core center $F=0.7426$; clock eigenvalue 0.170362 below vacuum 0.25.

Significance: The declared core density produces a spatially resolved clock mass well and a separately solved localized mode.

Limitation: Flat spherical branch only; neither gravity nor nonspherical perturbations follow from this profile.

bound-control

Question: Does the bound eigenvalue survive resolution controls and disappear without trapping?

Reading: Selected $E/Q=0.919412$; removing ν gives lowest eigenvalue 0.251711 versus 0.25 threshold.

Eigen error estimate $2.82e-07$.

Significance: The trapping coefficient is causally relevant to this bound mode in the registered family.

Limitation: A finite-box eigenvalue above 0.25 is a nonbound control. Radial Hessian diagnostics do not exclude all nonlinear or angular instabilities.

local-trace

Question: Does a local field trace tick for 100 periods when the unexcited core does not?

Reading: Nominal local trace completes 100.00 inferred signed cycles; local frequency 0.412749 versus eigenfrequency 0.41275. Unexcited trace is zero.

Significance: Unlike the stationary core density/projector, the field sampled inside the core has a repeatable local zero-crossing record.

Limitation: This is an ideal local field probe in the flat limit. Marker coupling, device noise and invariant reception comparison are future tests.

lifetime-convergence

Question: Does the mode persist with controlled charge, amplitude, grid and boundary errors?

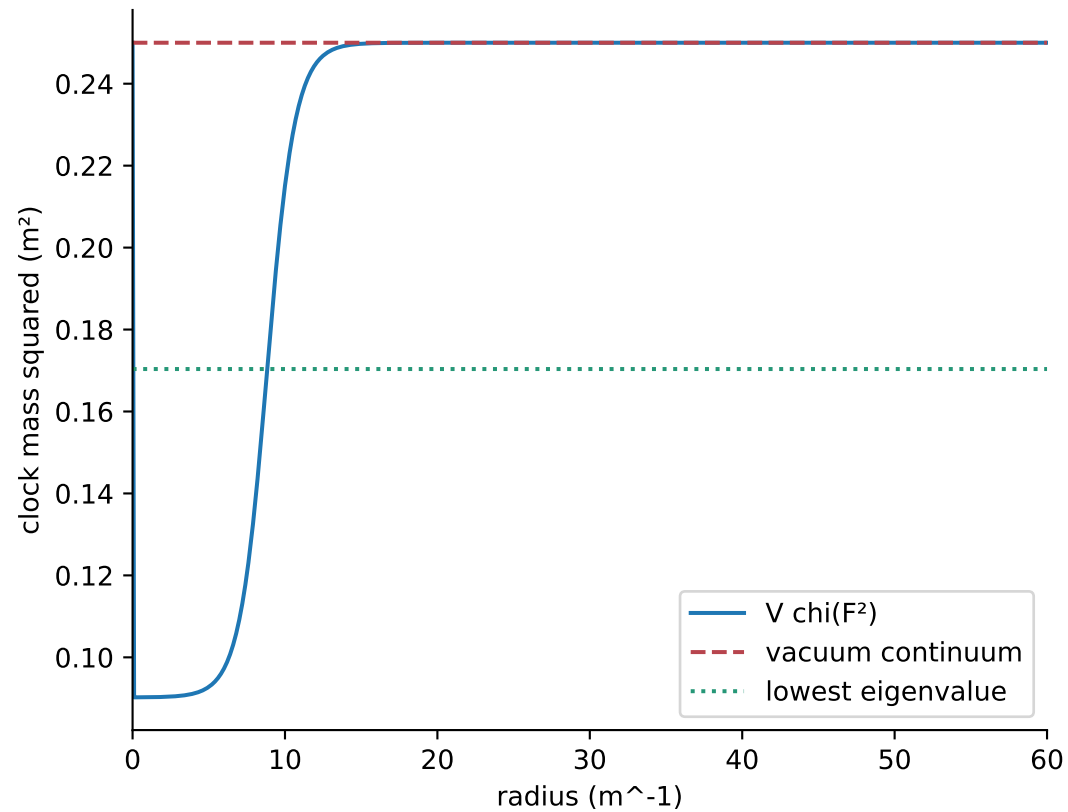
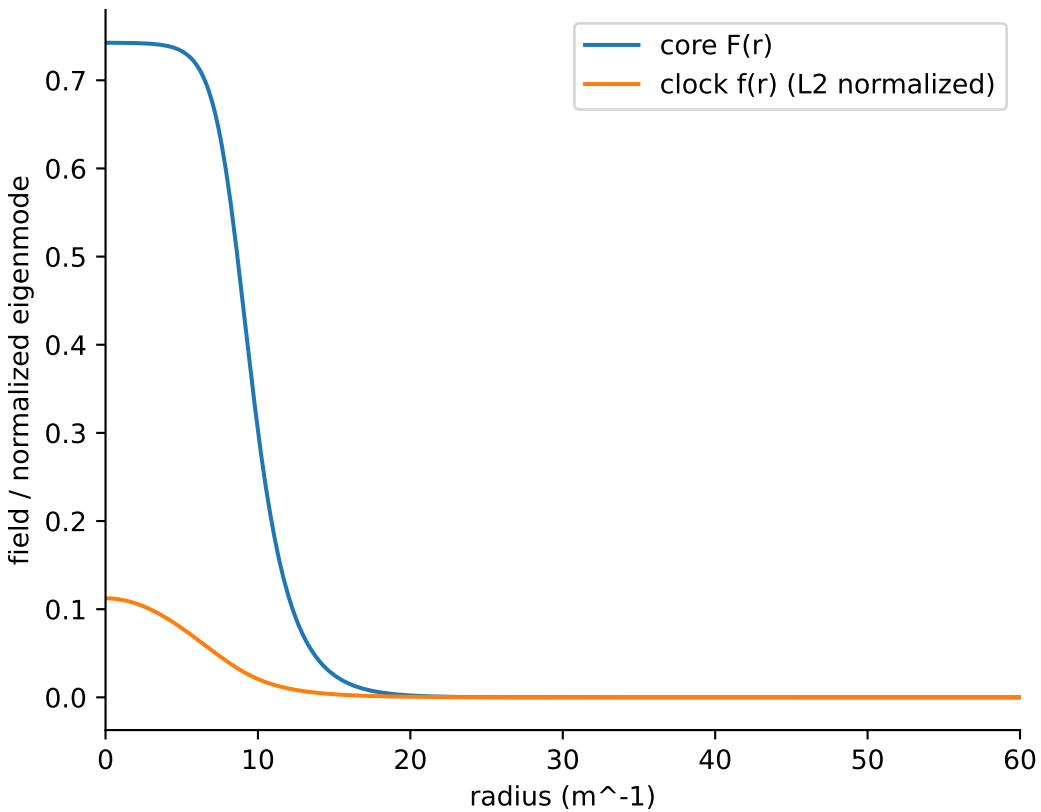
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Reading: Over 100 periods mode-energy loss $2.53\text{e-}05$; maximum charge ledger error $1.44\text{e-}15$; short-run frequency grid/domain difference $1.01\text{e-}05$.

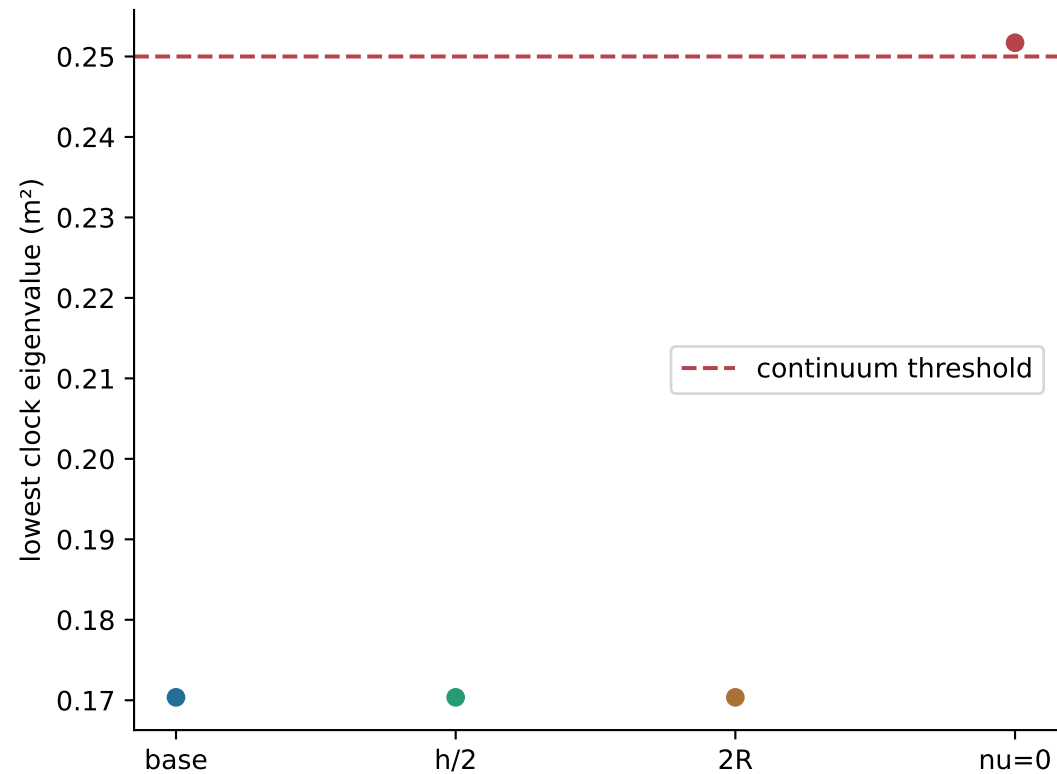
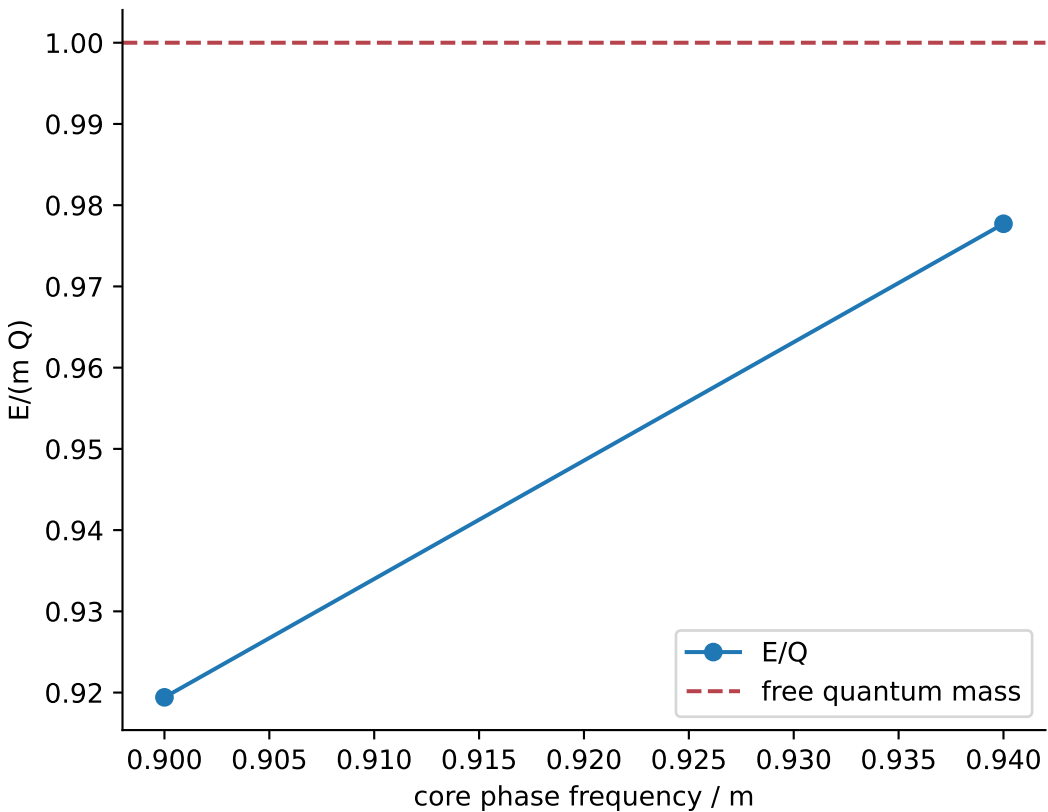
Significance: The duration and numerical controls quantify whether this core can provisionally serve as Test 7 calibration.

Limitation: Refined/wider runs and half/double amplitude controls cover 20 periods. An outer damping layer and untested nonspherical dynamics limit lifetime claims.

Spatial core and independently computed clock mode

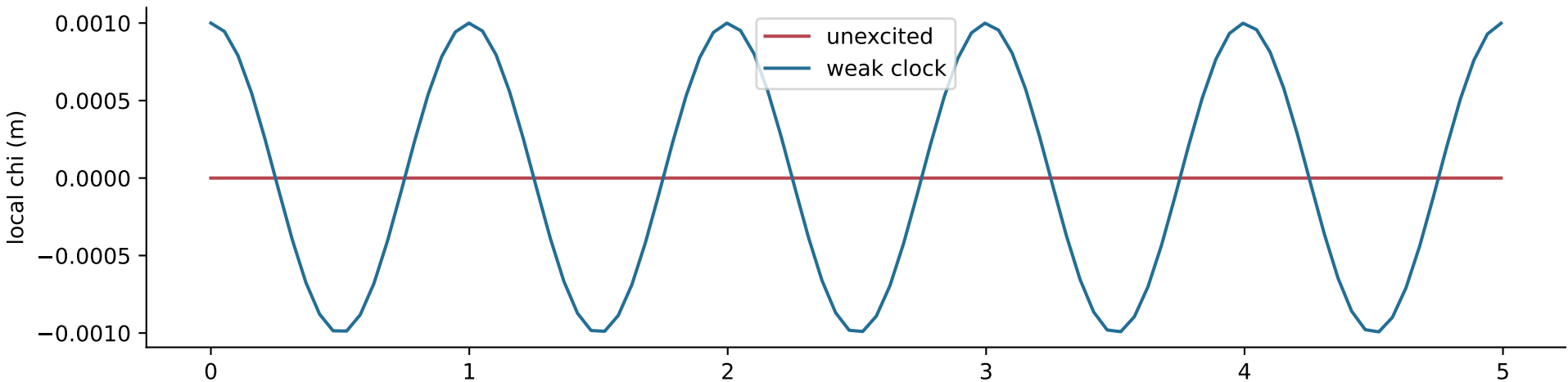


Core energetics and trapping failure control

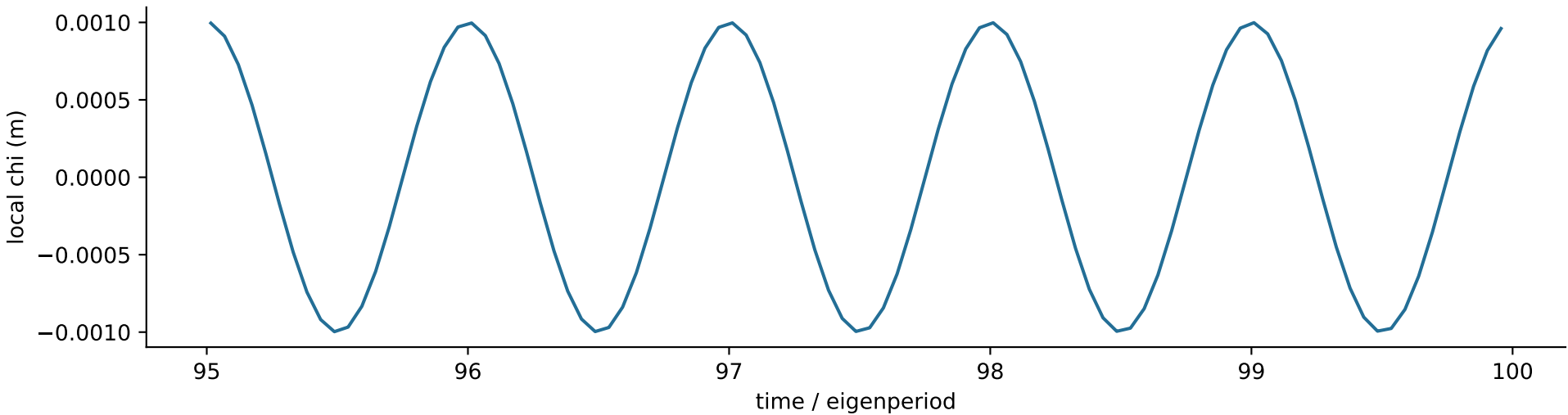


Local readout near core center, not a global fitted projection

first five periods

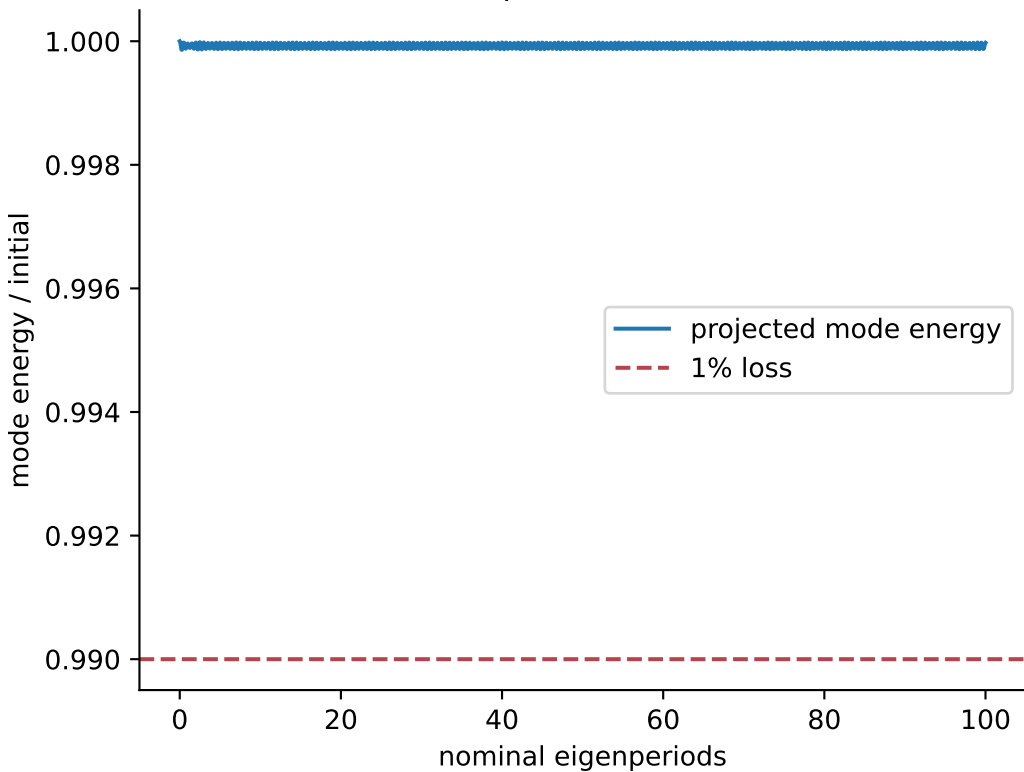


last five periods



Lifetime and charge; short amplitude/grid controls in exact saved data

100-period lifetime



conserved-charge ledger

